

Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains in Indian Railways: Architecture, Algorithmic Frameworks, and Operational Integration

Problem Landscape and Operational Context

The Indian Railways network constitutes one of the largest and most heavily saturated transportation systems in the world, spanning over 69,000 route kilometers, connecting 4,735 coaching stations across 17 operational zones, and handling over 23 million passengers every day. On critical trunk lines, particularly along the Golden Quadrilateral and High-Density Network (HDN) routes connecting major metropolitan centers, line capacity utilization regularly exceeds 100% to 140%. Coaching operations encompass a diverse fleet of services—ranging from semi-high-speed Vande Bharat expresses, Rajdhani, and Shatabdi services to long-distance Mail/Express trains, unreserved Jan Sadharan rakes, and dense suburban commuter networks. Managing punctual arrivals across this heterogeneous traffic mix presents substantial operational challenges, with average arrival delays exceeding 30 minutes, over 25% of coaching trains running more than 30 minutes behind schedule, and more than 7% experiencing cumulative delays beyond two and a half hours.

The legacy mechanism used by the National Train Enquiry System (NTES) and divisional control offices to estimate train arrivals relies heavily on static schedule extrapolation. This legacy approach records the latest manual or semi-automated station passing timestamp, adds the baseline scheduled section running time defined in the Working Time Table (WTT), and applies a static, lump-sum deduction based on built-in recovery margins.

This deterministic paradigm fails because train operations are governed by physical constraints, signaling interlocks, train precedence logic, temporary caution orders, and stochastic knock-on delays. A delay in railway operations is fundamentally bifurcated into primary and secondary disruptions. A primary delay arises from direct, exogenous physical disturbances such as traction power interruptions, overhead equipment (OHE) voltage fluctuations, rolling stock brake-binding, passenger emergency alarm chain pulling, mechanical track defects, or prolonged level crossing (LC) gate closures.

A secondary (reactionary or knock-on) delay occurs when this initial perturbation ripples through fixed track blocks, signaling circuits, and station throats, compelling trailing and opposing trains to decelerate, crawl, or halt at restrictive signal aspects. Because the legacy calculation assumes that a delayed train will immediately regain its scheduled path upon leaving a reporting station, it consistently overlooks downstream bottlenecks, conflicts with higher-priority traffic, and the cascading secondary delays that propagate across single-track and double-track corridors.

The inability to accurately forecast train arrivals generates systemic friction across multiple operational domains. Within station and yard environments, unexpected late arrivals cause

platform allocation conflicts, leading to yard-throat congestion where incoming rakes are forced to wait outside home signals, which in turn blocks the main lines for through-trains.

For rolling stock and maintenance logistics, erroneous ETAs disrupt pit-line maintenance schedules, coach watering cycles, mechanized internal cleaning, and onboard catering loading, which delays turnaround rake departures and creates new primary delays for outbound services.

In crew operations, unexpected running delays frequently cause locomotive pilots and train managers to exceed their statutory 10-hour Maximum Running Duty allowance under the Crew Management System (CMS), necessitating unscheduled mid-section relief stops that bring lines to a standstill.

For passengers, the disconnect between actual train positions and projected arrival times undermines public trust in digital information channels, causing missed intermodal transfers, congested platforms, and broader travel disruptions.

Operational Domain	Failure Modes of Legacy Static ETA	Impacted Systems and Infrastructure
Station & Platform Operations	Erroneous platform assignment, yard throat deadlocks, conflicting route-locking sequences, and unnecessary loop-line holding.	Control Office Application (COA), Route Relay Interlocking (RRI), Electronic Interlocking (EI).
Terminal & Depot Logistics	Disrupted pit-line maintenance allocations, delayed coach watering, cleaning bottlenecks, and late rake turnarounds.	Integrated Coaching Management System (ICMS), Coaching Yard Operations Management.
Locomotive & Crew Rostering	Violation of the 10-hour running duty rule, unplanned mid-section crew relief halts, and misaligned power power/loco balancing.	Crew Management System (CMS), Locomotive Link and Distribution Books.
Passenger & Intermodal Logistics	Platform crowding, missed rail and multi-modal transit connections, and inaccurate customer communications.	National Train Enquiry System (NTES), IRCTC Mobile Applications, Station Passenger Information Display Systems (PIDS).

Addressing these systemic challenges requires moving away from static schedule-based tracking toward a dynamic, data-driven ETA forecasting architecture. This platform must continuously ingest high-frequency satellite telematics, signaling interlocking states, infrastructure speed restrictions, and train interaction dynamics to generate probabilistic arrival forecasts that update as conditions change across the network.

Data Ingestion and Telematics Infrastructure

Developing an operational, dynamic prediction platform requires continuous, fault-tolerant telemetry ingestion across heterogeneous operational systems deployed across Indian Railways. The primary spatial tracking backbone is the Real-Time Train Information System (RTIS), developed by the Centre for Railway Information Systems (CRIS) in collaboration with the Indian Space Research Organisation (ISRO).

The RTIS onboard unit, installed inside the locomotive cab, integrates a dual-constellation positioning receiver powered by the GPS-Aided GEO Augmented Navigation (GAGAN) satellite system. GAGAN provides differential satellite corrections that yield meter-level spatial precision, preventing track-assignment ambiguity in complex multi-track corridors and station approach yards.

Telemetry transmission utilizes a dual-path communication setup:

- **Satellite Channel:** Telemetry is routed via an ISRO S-Band Mobile Satellite Service (MSS) uplink transponder on GSAT geostationary communication satellites, ensuring uninterrupted data flow across remote, desert, forest, and mountainous sections lacking terrestrial cellular connectivity.
- **Terrestrial Cellular Channel:** In urban centers, tunnels, and deep cuttings where satellite lines of sight can be obstructed, the device automatically falls back to multi-carrier 4G/LTE cellular modems.

The RTIS processing engine evaluates real-time coordinates against an embedded spatial geofence database of the rail network. The system transmits two primary data packet streams to the CRIS Central Location Server (CLS):

- **Periodic Telemetry:** Continuous mid-section positional reports—including latitude, longitude, instantaneous speed, and heading—transmitted every 30 seconds.
- **Event-Driven Telemetry:** Automated, geofence-triggered messages that log station arrivals, departures, run-through passings, and unexpected mid-section halts without requiring manual intervention from station masters or train drivers.

For diesel-hauled passenger and freight movements not yet covered by RTIS, the Remote Monitoring and Management of Locomotives and Trains (REMMLOT) framework provides cellular-based operational data into the CLS. Together, RTIS and REMMLOT supply live telemetry from more than 6,500 locomotives, with RTIS deployed across 2,700 electric locomotives in Phase-I and expanding to 6,000 additional locomotives across 50 sheds in Phase-II.

To produce accurate arrival forecasts, raw positional feeds must be integrated with data from multiple core enterprise platforms across the railway:

- **Control Office Application (COA):** The operational platform used across all divisional control offices. COA processes automated RTIS feeds from the CLS to plot digital control charts, capturing train routing, loop-line diversions, controller dispatch decisions, and planned line crossings.
- **National Train Enquiry System (NTES):** The centralized data distribution hub feeding passenger information channels, including public enquiry apps, SMS alerts, and web portals.
- **Integrated Coaching Management System (ICMS):** Maintains real-time operational status for coaching rakes, carriage compositions, punctuality records, and rolling stock maintenance allocations.
- **Crew Management System (CMS):** Tracks locomotive crew assignments, duty sign-on and sign-off records, and mandatory rest periods, identifying trains at risk of unscheduled stops due to duty-time limits.
- **Working Time Table (WTT):** Contains foundational physical running baselines, such as Booked Speeds, Maximum Permissible Speeds (MPS), Normal Sectional Running Times (NSRT), and Minimum Sectional Running Times (MSRT).
- **Caution Order Information Management System (COIMS / e-TSR):** Provides live records of Temporary Speed Restrictions (TSRs) stemming from track maintenance, bridge inspections, civil works, or rail replacement activities.

- **Trackside Signaling and Interlocking Telemetry:** Data-logger logs and Electronic Interlocking (EI) records providing real-time signal aspect states (Green, Double Yellow, Yellow, Red) and route alignments.
- **Meteorological and Low-Visibility Telemetry:** Station Visibility Test Object (VTO) records, Fog-PASS unit activation alerts, and weather feeds from the India Meteorological Department (IMD).

Infrastructure Source	Protocol and Ingestion Latency	Update Frequency	Data Granularity	Core Feature Contribution
RTIS Locomotive Cab Units	MQTT / CoAP over ISRO S-Band Satellite & 4G/LTE; < 2 s.	Every 30 seconds (periodic) plus geofence event triggers.	Geodetic coordinates, speed, heading, locomotive ID.	Real-time position tracking, instantaneous speed, and mid-section halt identification.
COA Control Charts	Apache Kafka / CDC via Enterprise Relational DB; < 1 s.	Event-driven upon dispatch action.	Section occupancies, platform assignments, train meets.	Downstream route reservations, track priority, and passing maneuvers.
Electronic Interlocking (EI)	Modbus over TCP / Serial-to-Ethernet; < 5 s.	Continuous state streaming.	Signal aspects (Red, Yellow, Green), point switch alignments.	Micro-scale signal aspect restrictions and block clearance speeds.
Working Time Table (WTT)	Batch ETL via PostgreSQL; Weekly/Seasonal update.	Static database updates.	Station-to-station MSRT, NSRT, booked speed limits.	Baseline operational physics and scheduled dwell tolerances.
e-TSR Caution Orders	RESTful JSON Push; Hourly or on-demand.	Real-time bulletin publishing.	Kilometric start/end markers, temporary speed caps.	Dynamic acceleration and running-time penalties across sections.
Fog-PASS & Weather Feeds	GeoJSON / REST over HTTPS; 5 to 15 min.	Continuous monitoring intervals.	Visibility range (meters), fog boundaries, precipitation.	Enforcement of fog-related speed restrictions and altered headways.

Spatio-Temporal Algorithmic Modeling Framework

Graph Formulation of the Railway Network

Railway networks cannot be modeled accurately using standard Euclidean spatial techniques because train movements are constrained to physical tracks and governed by discrete block signaling and interlocking switches. Instead, the rail system is structured as a directed, dynamic multi-graph:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{W}_s, \mathcal{W}_t)$$

where:

- $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ represents the set of N nodes, corresponding to physical block stations, intermediate junction cabins, platform tracks, and automatic signaling block divisions.
- $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represents directed edges that correspond to physical track sections connecting contiguous nodes.
- $\mathcal{W}_s$ represents static topological edge features, including track count (single, double, triple, or quadruple lines), edge length L_{ij} , ruling gradients, curve geometry, switch point turnout angles, and Maximum Permissible Speed $V_{ij}^{\max}$.
- $\mathcal{W}_t$ represents dynamic operational edge features evaluated at time t , including current signal aspect states, active temporary speed restrictions, track occupancy counts, and localized visibility indices.

Train trajectories traversing this graph are represented as discrete sequences of operational events. For any train k , its progression across a sequence of station nodes is defined by actual arrival times a_i^k , departure times d_i^k , and delay states $\delta_i^k = a_i^k - s_i^k$, where s_i^k denotes the scheduled arrival time published in the official timetable.

Hierarchical Prediction Architecture

A single machine learning model cannot effectively capture both large-scale, network-wide delay diffusion and the localized, physics-based dynamics of an individual train. The operational forecasting framework therefore uses a two-tier hierarchical structure:

- **Tier 1 (Macro-Scale Network Level):** A Spatio-Temporal Graph Neural Network (STGNN) models network-wide congestion, headway compression, and delay diffusion across interconnected corridors.
- **Tier 2 (Micro-Scale Train Level):** An ensemble combining Gradient Boosted Decision Trees (LightGBM) with Temporal Convolutional Networks (TCN) or Bidirectional LSTMs predicts specific travel times for individual trains, conditioned on the spatial embeddings generated by Tier 1.

At the macro scale, the system implements a Railway-centric Spatio-Temporal Graph Convolutional Network (RSTGCN) designed to capture operational conditions across large-scale rail systems. The RSTGCN architecture incorporates train-frequency-aware spatial attention mechanisms to model interactions between stations and dynamic capacity bottlenecks. Given a dynamic node feature matrix $\mathbf{X}_t \in \mathbb{R}^{N \times F}$, where each node v_i tracks F operational features (such as average inbound arrival delay, departure delay, total backlog, and active hourly headway), spatial graph convolutions capture dependencies across neighboring stations. The spatial attention weight α_{ij}^t between station node v_i and adjacent station $v_j \in \mathcal{N}(v_i)$ dynamically balances track connectivity against real-time scheduled train frequencies:

$$\alpha_{ij}^t = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}^T [\mathbf{W} \mathbf{x}_i^t \parallel \mathbf{W} \mathbf{x}_j^t \parallel \mathbf{W}_f f_{ij}^t]\right)\right)}{\sum_{m \in \mathcal{N}(v_i)} \exp\left(\text{LeakyReLU}\left(\mathbf{a}^T [\mathbf{W} \mathbf{x}_i^t \parallel \mathbf{W} \mathbf{x}_m^t \parallel \mathbf{W}_f f_{im}^t]\right)\right)}$$

where:

- $\mathbf{W} \in \mathbb{R}^{F' \times F}$ is a shared linear projection matrix.
- f_{ij}^t represents the scheduled train frequency (active train count per hour) moving between nodes v_i and v_j during time window t .
- $\mathbf{a} \in \mathbb{R}^{2F' + F_f}$ is a learnable attention parameter vector, and $\parallel$ denotes concatenation.

denotes vector concatenation.

Temporal dependencies across historical observation horizons $\{t-T+1, \dots, t\}$ are processed using gated Temporal Convolutional Networks (TCN) with dilated causal convolutions or Bidirectional GRU layers. This design captures both non-linear delay accumulation and downstream recovery trends, updating global station delay embeddings across the network.

Delay Propagation Dynamics

Delay propagation between two consecutive stations v_i and v_j for train k depends on scheduled travel time, minimum physical running time, and headway buffer consumption:

$$\Delta_j^k = \max\left(0, \Delta_i^k + \Delta t_{\text{actual}}^k(i, j) - \Delta t_{\text{sched}}^k(i, j) - \beta_{ij}^k\right) + \sum_{m \in \mathcal{P}_k(i, j)} \gamma_m \left(h_{\min} - h_{mk}\right)^+$$

where:

- Δ_i^k is the departure delay of train k at station v_i .
- $\Delta t_{\text{actual}}^k(i, j)$ and $\Delta t_{\text{sched}}^k(i, j)$ represent the actual and scheduled running times between nodes v_i and v_j .
- β_{ij}^k represents the schedule buffer (Engineering Allowance and Traffic Recovery Time) allocated for train k within section (i, j) .
- $\mathcal{P}_k(i, j)$ is the set of preceding or conflicting trains sharing the track block or station platform throat ahead of train k .
- h_{mk} is the operational headway (temporal separation) between preceding train m and train k , while $h_{\min}$ is the absolute minimum safety signaling headway.
- $\gamma_m \in [0, 1]$ is the precedence interaction penalty determined by train priority hierarchies. The notation $(z)^+ = \max(0, z)$ ensures that headway penalties apply only when real headway compresses below safety minimums ($h_{mk} < h_{\min}$).

Micro-Scale Run-Time Prediction and Uncertainty Quantification

At Tier 2, the pipeline extracts the localized graph embedding $\mathbf{h}_{v_j}^t$ from the Tier 1 RSTGCN and combines it with train-specific tabular features, including rolling stock class, rake length, gross trailing tonnage, locomotive horsepower, driver performance indices, and active caution orders. LightGBM models predict running times across downstream sections due to their high execution speed, leaf-wise tree splitting efficiency, and native handling of high-cardinality categorical variables such as Train Number and Station Code.

Because deterministic point predictions cannot communicate operational risk during service disruptions, the Tier 2 models generate probabilistic prediction intervals via **Quantile**

Regression optimized with the Pinball Loss function:

$$\mathcal{L}_{\tau}(y, \hat{y}_{\tau}) = \sum_i: y_i \geq \hat{y}_{\tau, i} \tau |y_i - \hat{y}_{\tau, i}| + \sum_i: y_i < \hat{y}_{\tau, i} (1 - \tau) |y_i - \hat{y}_{\tau, i}|$$

The model is trained across three quantiles: $\tau \in \{0.10, 0.50, 0.90\}$. The median forecast

$\hat{y}_{0.50}$ provides the expected point arrival time, while $\hat{y}_{0.10}$ and $\hat{y}_{0.90}$ establish an initial 80% confidence interval.

To resolve empirical under-coverage and deliver reliable finite-sample coverage guarantees,

raw quantile outputs are calibrated using **Conformalized Quantile Regression (CQR)** combined with **Mondrian Conformal Prediction (MCP)**. Using a holdout calibration dataset $(X_i, Y_i)_{i=1}^{n_{\text{cal}}}$, nonconformity scores E_i are computed:

$$E_i = \max\left(\hat{q}_{\alpha/2}(X_i) - Y_i, Y_i - \hat{q}_{1-\alpha/2}(X_i)\right)$$

The $(1 - \alpha)(1 + 1/n_{\text{cal}})$ -th empirical quantile of the nonconformity scores $\{E_1, \dots,$

$E_{n_{\text{cal}}}\}$, denoted as $Q_{1-\alpha}(E)$, is calculated. The final calibrated ETA prediction interval is defined as:

$$C(X_{n+1}) = \left[\hat{q}_{\alpha/2}(X_{n+1}) - Q_{1-\alpha}(E), \hat{q}_{1-\alpha/2}(X_{n+1}) + Q_{1-\alpha}(E) \right]$$

Mondrian stratification groups calibration samples across service tiers (Premium Express, Mail/Express, Ordinary Passenger) and weather regimes (Clear vs. Foggy), guaranteeing valid conditional coverage across operational edge cases.

Operational Constraints and Domain Rules Integration

Working Time Table Mechanics and Allowance Consumption

The Working Time Table (WTT) balances physical line capacity against rolling stock capabilities. Train running durations are structured using two operational metrics:

- **Minimum Sectional Running Time (MSRT):** The physical transit time achievable when a train runs at the Maximum Permissible Speed (MPS) of the section, constrained solely by trailing tonnage, tractive effort, curvature, and ruling gradients.
- **Normal Sectional Running Time (NSRT):** Calculated using the lower Booked Speed, creating an operational cushion that loco pilots can use to absorb minor running variations.

To manage operational disruptions, railway timetables incorporate two structural time allowances:

- **Engineering Allowance (EA):** Time budgeted (typically 2 to 6 minutes per 100 km) to absorb the deceleration, slow running, and acceleration phases required by temporary caution orders, track renewal sites, and maintenance blocks.
- **Traffic Recovery Time (TRT):** Buffer time placed upstream of major junction approaches and terminal stations, allowing late-running trains to recover schedule before entering congested station throats.

The static ETA system assumes that downstream recovery buffers will be fully realized, leading to inaccurate predictions when trains run on schedule or experience minor delays. The dynamic prediction engine monitors the real-time status of TSRs on track sections: if no caution orders are active in a section, the model credits the Engineering Allowance as recoverable time; if severe TSRs are active, the model applies calculated speed penalties, dynamically modifying the downstream ETA.

Precedence Rules and Priority Matrices

When multiple trains contend for shared track capacity, section controllers make dispatch decisions based on the operational hierarchy codified in Indian Railways operating manuals. Premium coaching trains (Vande Bharat, Rajdhani, Shatabdi) hold absolute precedence over

standard Superfast and Mail/Express services, which take priority over suburban passenger rakes, while freight services operate within remaining line capacity.

On double-track lines, if a lower-priority train runs ahead of a delayed higher-priority service, dispatchers divert the lower-priority train into an intermediate station loop line to allow the higher-priority service to overtake on the main through-line. This diversion imposes a significant, non-linear time penalty on the lower-priority service:

- Decelerating from sectional speed to turnout speed (typically 15 km/h or 30 km/h) over switch points into the loop line.
- Halting on the loop line while the higher-priority train traverses the station and clears downstream braking blocks.
- Accelerating from a dead stop back to sectional running speed once points and signals are reset.

The Tier 1 graph model tracks this interaction continuously: as the spatial separation between approaching trains decreases, the model calculates the probability of a loop-line diversion for the lower-priority service and adjusts its downstream ETA to account for deceleration, dwell, and re-acceleration penalties.

Low-Visibility Operations and Fog Management Protocols

During northern India's winter months (December through February), dense radiation fog severely impairs signal sighting across Northern Railway, North Central Railway, and East Central Railway divisions. When station staff cannot visually confirm the station Visibility Test Object (VTO) at a distance of 180 meters, low-visibility operating rules are triggered.

Under these low-visibility conditions:

- Loco pilots operate using GPS-based Fog-PASS (Fog Pilot Assistance System for Safety) units, which provide audio-visual alerts as trains approach landmarks, warning boards, level crossings, and signals within a pre-configured geofence.
- Train speeds are capped at 60 km/h to 75 km/h, regardless of whether the track infrastructure normally supports 110 km/h or 130 km/h running.
- On routes equipped with Automatic Block Signaling (ABS), modified rules restrict train density to a maximum of two trains between adjacent stations, reducing theoretical line capacity by roughly 50%.

The dynamic ETA engine directly incorporates these operational rules. When Fog-PASS activation flags or VTO declarations are logged in an operational division, the model updates the maximum permissible velocity $V_{ij}^{\max}$ and adjusts block clearance headways, generating realistic arrival bounds rather than projecting clear-weather running profiles.

Feature or Operational Aspect	Legacy Static Schedule Approach	Dynamic Machine Learning Architecture
Locomotive Position Updates	Manual station master logs or intermittent NTES status polls.	Continuous 30-second RTIS satellite updates and automated geofence event triggers.
Schedule Buffer Handling	Static, lump-sum TRT and EA deductions applied regardless of section status.	Dynamic allowance adjustments based on live caution orders and maintenance blocks.
Secondary Delay Modeling	Assumes zero secondary delay	Graph-attention modeling of

Feature or Operational Aspect	Legacy Static Schedule Approach	Dynamic Machine Learning Architecture
	propagation across neighboring trains.	secondary knock-on delays and headway compression.
Train Precedence Dynamics	Projects conflicting train paths as independent, uninterrupted runs.	Dynamic modeling of loop-line holding penalties and turnout deceleration.
Adverse Weather Response	Unadjusted timetable calculations until trains experience compounding delays.	Automated speed caps (60–75 km/h) and doubled headway buffers in active fog zones.
Forecast Output Format	Deterministic point estimate (frequently inaccurate).	Calibrated probabilistic arrival windows with explicit confidence bounds.

Real-Time System Architecture and Serving Pipeline

Deploying dynamic ETA forecasting across thousands of concurrently operating coaching trains requires a distributed, low-latency streaming and inference pipeline designed for high availability and fault tolerance.

The ingestion tier captures high-frequency RTIS positional packets via an enterprise Apache Kafka cluster deployed across multi-zone datacenters. Streaming telemetry is ingested by an Apache Flink processing cluster, which executes event-time sliding window aggregations across stateful operators.

This streaming layer handles several key feature engineering tasks:

- Calculating instantaneous section acceleration and running speed profiles.
- Identifying mid-section halts exceeding 120 seconds outside scheduled timetables.
- Computing real-time spatial headway to the immediate preceding train on the same track circuit.
- Matching coordinates against spatial track vectors to verify mainline vs. loop line diversions.

Engineered feature vectors are written to an in-memory Feast feature store supported by Redis Enterprise, providing sub-millisecond retrieval during real-time inference.

Model serving is orchestrated through NVIDIA Triton Inference Server running the ONNX Runtime. Triton executes dynamic model batching, grouping incoming inference requests across multiple trains into parallel tensor operations executed on enterprise GPUs.

The inference workflow operates in two phases:

1. Every 60 seconds, or upon significant track disruption, the Tier 1 RSTGCN model updates global node embeddings across the 4,735 stations of the network graph.
2. For each individual train, the Tier 2 LightGBM and TCN quantile regressor evaluates running times to upcoming stations using live feature vectors from Redis combined with current Tier 1 embeddings.
3. Conformal prediction bounds are calculated, outputting expected arrival times along with the 10th and 90th percentile bounds.
4. Triton delivers these predictions across the entire active network (~10,000 active coaching and freight trains) within a 50-millisecond inference window.

Downstream systems consume predictions from an Envoy-managed API gateway and Redis edge cache. Divisional train controllers receive dynamic look-ahead trajectories directly on their

COA electronic control charts via high-throughput gRPC connections, supporting proactive dispatch and precedence planning.

Concurrently, public-facing applications (the NTES passenger mobile app, the IRCTC booking portal, and station Passenger Information Display Systems) consume predictions via scalable WebSockets and REST APIs, providing travelers with updated arrival windows and explicit confidence margins.

Architectural Layer	Core Technologies	Functional Responsibilities	Latency & Performance SLA
Ingestion Pipeline	Apache Kafka, ISRO Satcom Gateway.	Ingestion of raw GAGAN/GPS telematics and external feeds.	Latency < 2 seconds end-to-end.
Stream Analytics	Apache Flink, Feast Feature Store.	Stateful feature transformation and fast key-value lookups.	Window processing latency < 100 ms.
Inference Engine	NVIDIA Triton, ONNX Runtime.	Parallel evaluation of RSTGCN and LightGBM quantile models.	Batch inference execution < 50 ms.
Caching Layer	Redis Enterprise Cluster.	In-memory storage of downstream station ETAs and confidence bounds.	Data retrieval latency < 5 ms.
<i>API Delivery</i>	Envoy Proxy Gateway, WebSockets.	Secure routing to NTES apps, COA consoles, and platform displays.	Client response latency < 200 ms.

Model Evaluation, MLOps, and Network Scalability

Evaluation Framework and Empirical Benchmarks

Model accuracy is evaluated across both deterministic point estimates and probabilistic interval calibration.

Deterministic accuracy is measured using standard regression metrics:

- **Mean Absolute Error (MAE):** Measures average prediction error in absolute minutes:

$$\text{MAE} = \frac{1}{M} \sum_{k=1}^M |y_k - \hat{y}_k|$$
- **Root Mean Squared Error (RMSE):** Penalizes severe outlier errors, reflecting major unpredicted disruptions:

$$\text{RMSE} = \sqrt{\frac{1}{M} \sum_{k=1}^M (y_k - \hat{y}_k)^2}$$
- **Mean Absolute Percentage Error (MAPE):** Normalizes delay relative to scheduled running duration:

$$\text{MAPE} = \frac{100\%}{M} \sum_{k=1}^M \left| \frac{y_k - \hat{y}_k}{y_k} \right|$$

Evaluations conducted on comprehensive datasets covering 4,735 stations across all 17 zones of the Indian Railways demonstrate that the RSTGCN model outperforms baseline methods—including historical averages, gradient boosting without graph features, and standard STGCN architectures—achieving reductions of up to 18% in MAE, 14% in MAPE, and 8% in RMSE.

Probabilistic predictions are evaluated using uncertainty metrics:

- **Prediction Interval Coverage Probability (PICP):** Validates that the true arrival time falls within the predicted interval $[\hat{y}_{\text{lower}}, \hat{y}_{\text{upper}}]$ at the nominal level (e.g., $\geq 80\%$):

$$\text{PICP} = \frac{1}{M} \sum_{k=1}^M \mathbb{I}(\hat{y}_k \in [\hat{y}_{\text{lower}}, \hat{y}_{\text{upper}}])$$
- **Mean Prediction Interval Width (MPIW):** Measures the sharpness of the prediction band:

$$\text{MPIW} = \frac{1}{M} \sum_{k=1}^M (\hat{y}_{\text{upper}, k} - \hat{y}_{\text{lower}, k})$$
- **Continuous Ranked Probability Score (CRPS):** Measures probabilistic accuracy across the entire distribution.

Edge-Case Handling and Anomaly Recovery

Operating telematics across a large-scale railway network requires handling several common data anomalies:

- **The Midnight Rollover Anomaly:** Timetable processing frequently misinterprets arrival times immediately following midnight (00:01), calculating artificial 24-hour (1440 minutes) delays. Data pipelines apply cyclic modulo transformations to normalize date transitions across cross-midnight runs.
- **Satellite Dropout & Signal Shadow:** When trains traverse deep tunnels, urban gorges, or dense electrified catenary networks where GPS/GAGAN signals degrade, the pipeline transitions to a continuous Extended Kalman Filter (EKF). The EKF fuses the last known satellite coordinate with locomotive wheel tachometer pulses and trackside transponder pings, maintaining accurate position tracking until satellite visibility is restored.
- **Station Loop Dwell Ambiguity:** When a train halts on a loop line to allow an overtake, naive models may mistake the stop for an unscheduled mechanical breakdown. The pipeline correlates track circuit occupancy flags with the positions of adjacent higher-priority trains to classify the halt as an operational overtake, dynamically predicting line clearance based on the preceding train's clearance time.

Continuous Learning, Drift Detection, and Scalability

Railway networks experience operational drift driven by seasonal timetable adjustments, summer passenger surges, and winter weather patterns. In addition, ongoing infrastructure projects—such as track doubling, corridor electrification, the operationalization of Dedicated Freight Corridors (DFCs), and the deployment of Kavach Automatic Train Protection (ATP)—continuously alter baseline network capacities and minimum headways.

To account for these evolving conditions, the MLOps pipeline implements a continuous monitoring and updating framework:

- **Drift Detection:** The pipeline tracks Population Stability Index (PSI) and Wasserstein Distance across input features (such as section transit times and headway variances) and prediction residuals.
- **Retraining Cadence:** Tier 2 LightGBM regressors are retrained weekly using sliding-window operational logs. The Tier 1 RSTGCN model undergoes monthly retraining to incorporate broader network topology changes.
- **Shadow Deployment:** Newly trained models run in shadow mode for 72 hours alongside production models. Candidate models are promoted to production via canary routing only after demonstrating a $\geq 5\%$ improvement in MAE while maintaining required conformal

prediction coverage thresholds.

Strategic Implementation Roadmap

Transforming Indian Railways' arrival forecasting requires a phased rollout aligned with technological and infrastructural readiness across operational zones.

Phase 1: High-Density Trunk Corridors (Months 1-12)

Delhi-Mumbai & Delhi-Howrah Routes | RTIS Phase-I/II Ingestion | Dual-Tier Inference Engine



Phase 2: Network-Wide

Zonal Expansion (Months 13-24)

Rollout to All 17 Operational Zones | Caution Orders (e-TSR) | Fog-PASS & Weather Ingestion



Phase 3: Autonomous Closed-Loop

Integration (Months 25-36)

Kavach SIL-4 ATP Telemetry | Automated Dispatch Advisories | Real-Time Schedule Rescheduling

Phase 1: High-Density Trunk Corridors (Months 1 to 12)

The initial deployment targets high-density routes along the Golden Quadrilateral, focusing on the Delhi-Howrah and Delhi-Mumbai trunk corridors. The architecture ingests live data from RTIS Phase-I installations (2,700 locomotives) and initial Phase-II rollouts through the CRIS Central Location Server. The Tier 1 RSTGCN network graph is calibrated on pilot divisions, integrating directly with electronic control charts in the Control Office Application (COA) to validate model accuracy against actual dispatch patterns.

Phase 2: Network-Wide Zonal Expansion (Months 13 to 24)

The platform expands across all 17 operational zones, incorporating 4,735 coaching stations into the spatio-temporal graph. The ingestion layer integrates live feeds from the Caution Order Information Management System (e-TSR), Fog-PASS geofencing alerts, and meteorological feeds. APIs are deployed to support public and internal channels, delivering dynamic arrival windows and confidence bounds to the NTES passenger application, the IRCTC portal, and station display systems.

Phase 3: Autonomous Closed-Loop Integration (Months 25 to 36)

The forecasting platform integrates bidirectionally with Kavach, Indian Railways' indigenous SIL-4 Automatic Train Protection system. Ingesting real-time cab-signaling states and trackside

transponder feeds enables the system to support computerized traffic management, generating automated dispatch advisories and dynamic conflict-resolution recommendations for section controllers. This operational integration shifts railway management from reactive delay tracking to proactive network optimization, improving overall punctuality, line capacity utilization, and passenger satisfaction across the national network.

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